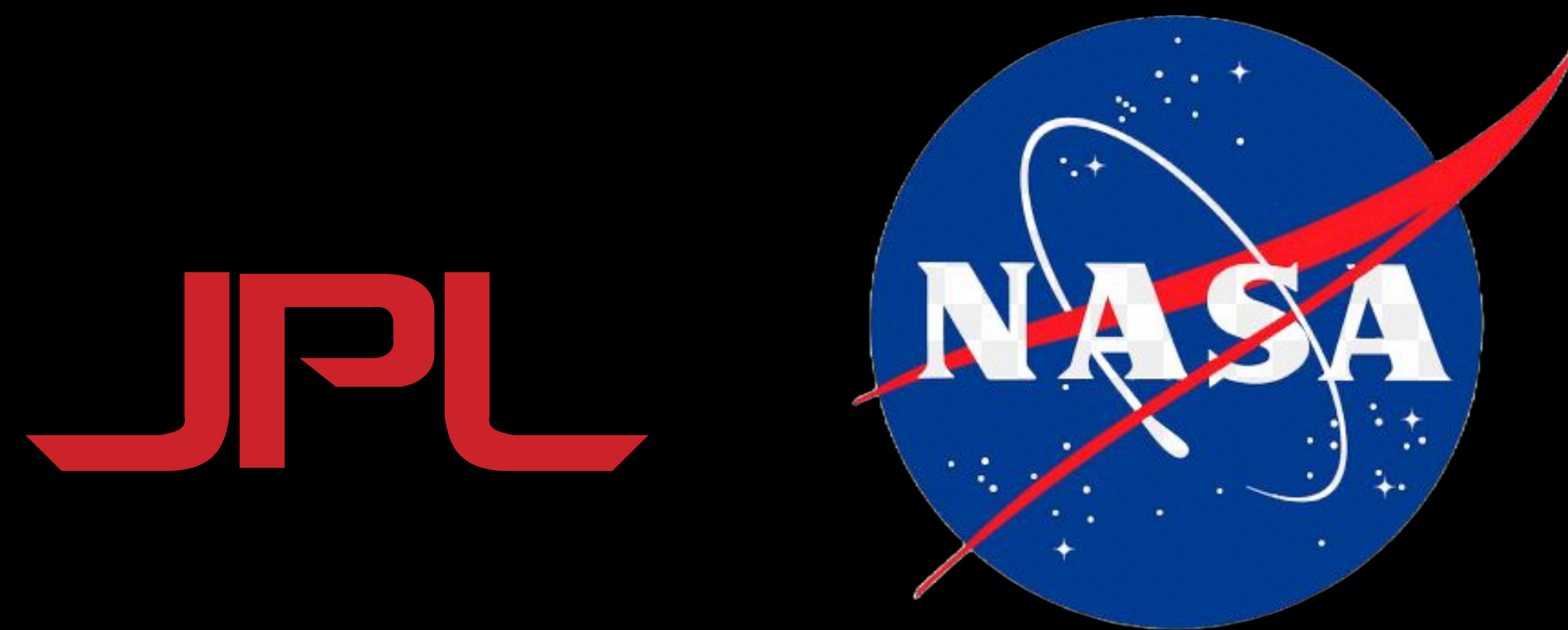




Mars Hard Lander

Abdulrahman Almutairi, Ahmad Sai, Anthony Hamamji, Nasser Al Muaili, Conner Neuhart, Bilim Sydykov



Features

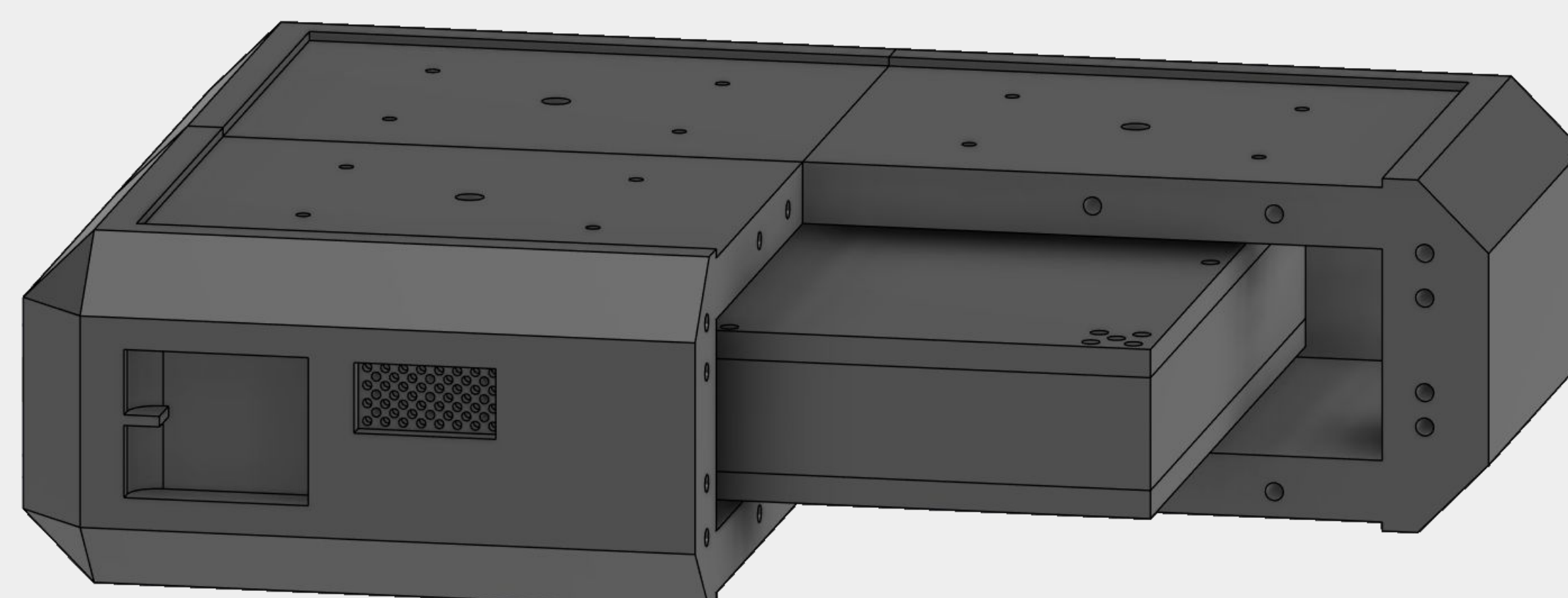
- Multi-sensor module
- Wireless Data Transmission
- Batteries last at least 2 Sols
- Up to 4 modular payload ports
- Onboard data storage
- Temperature Controlled Enclosure

Measurements

- Internal and External Temperature
- Humidity
- Gyroscope
- Landing Acceleration
- Voltage Monitoring

Enclosure Design

Enclosure is built mainly out of TPU, using polyethylene foam as internal insulation with layers of nylon and mylar as external insulation to trap heat and keep out cold Mars temperatures



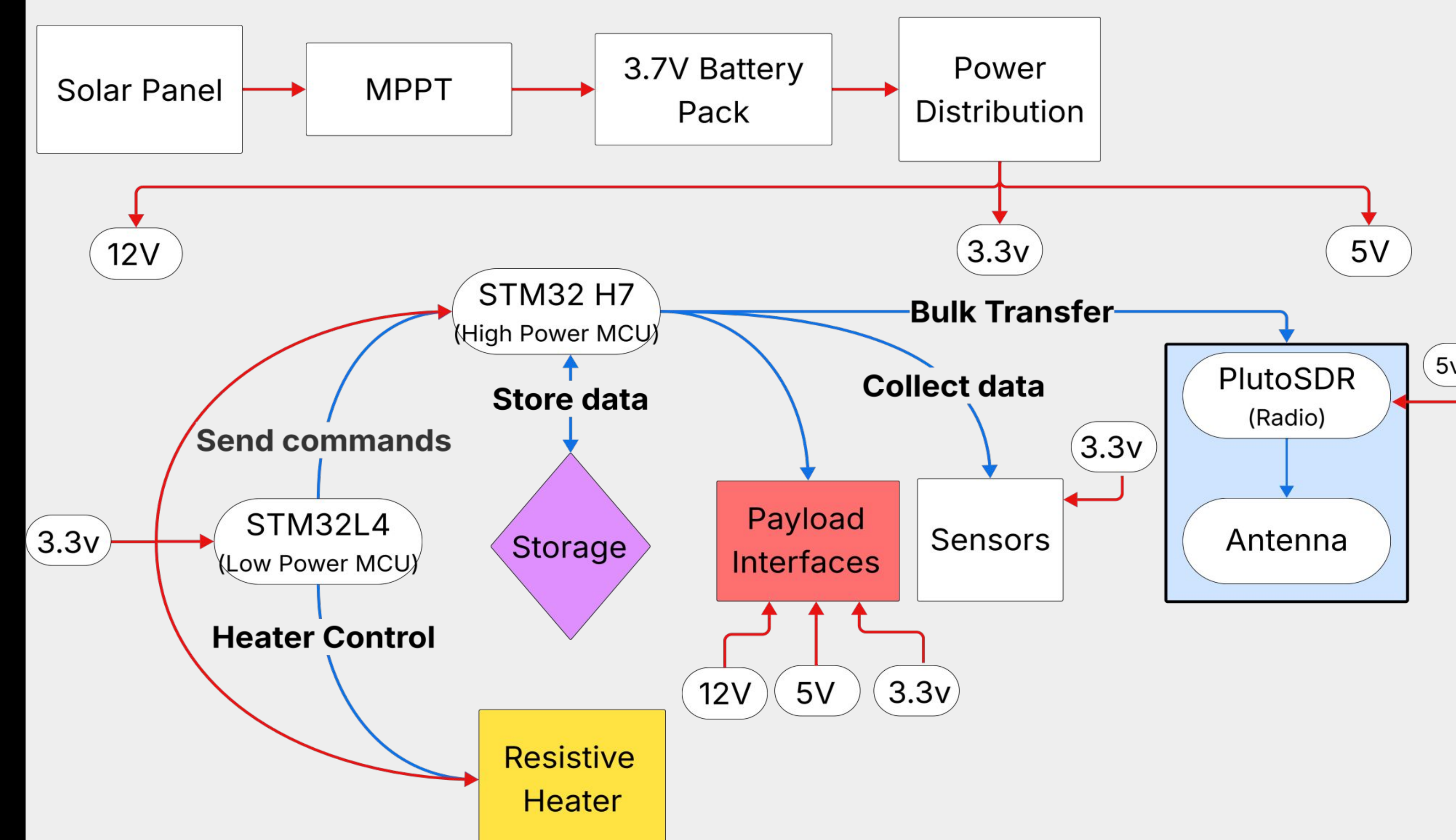
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JPL Advisor: Steve Matousek
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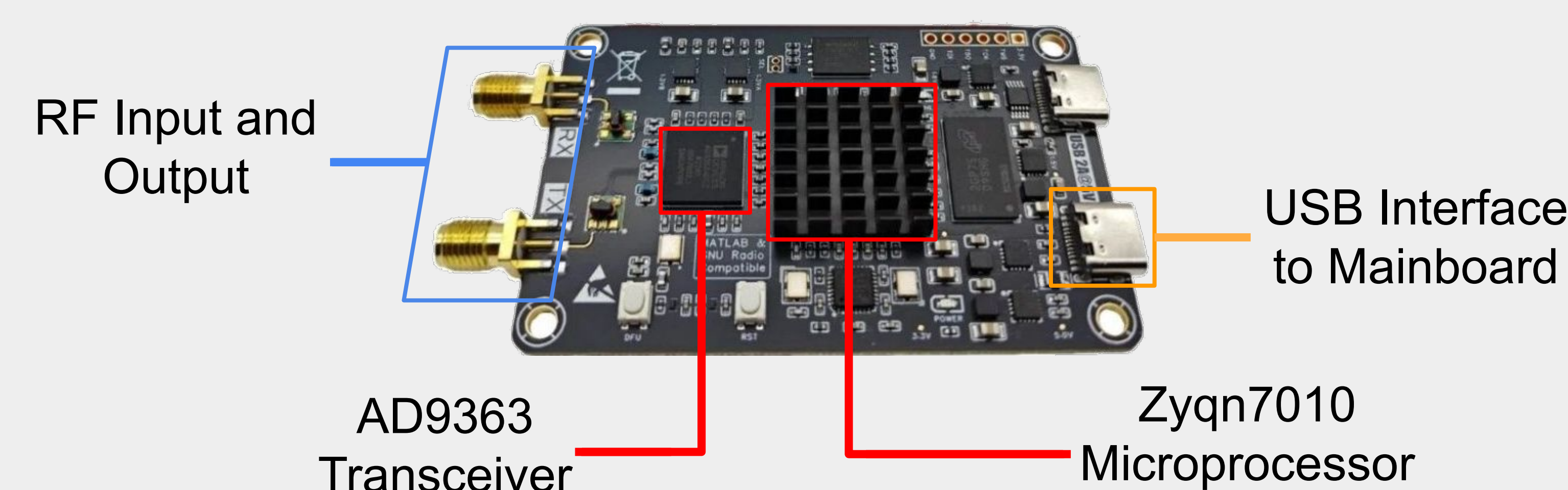
Overview

A low cost hard lander electrical system with temperature regulation, modular payloads, wireless communication, and reliable data handling for harsh environments

System Diagram

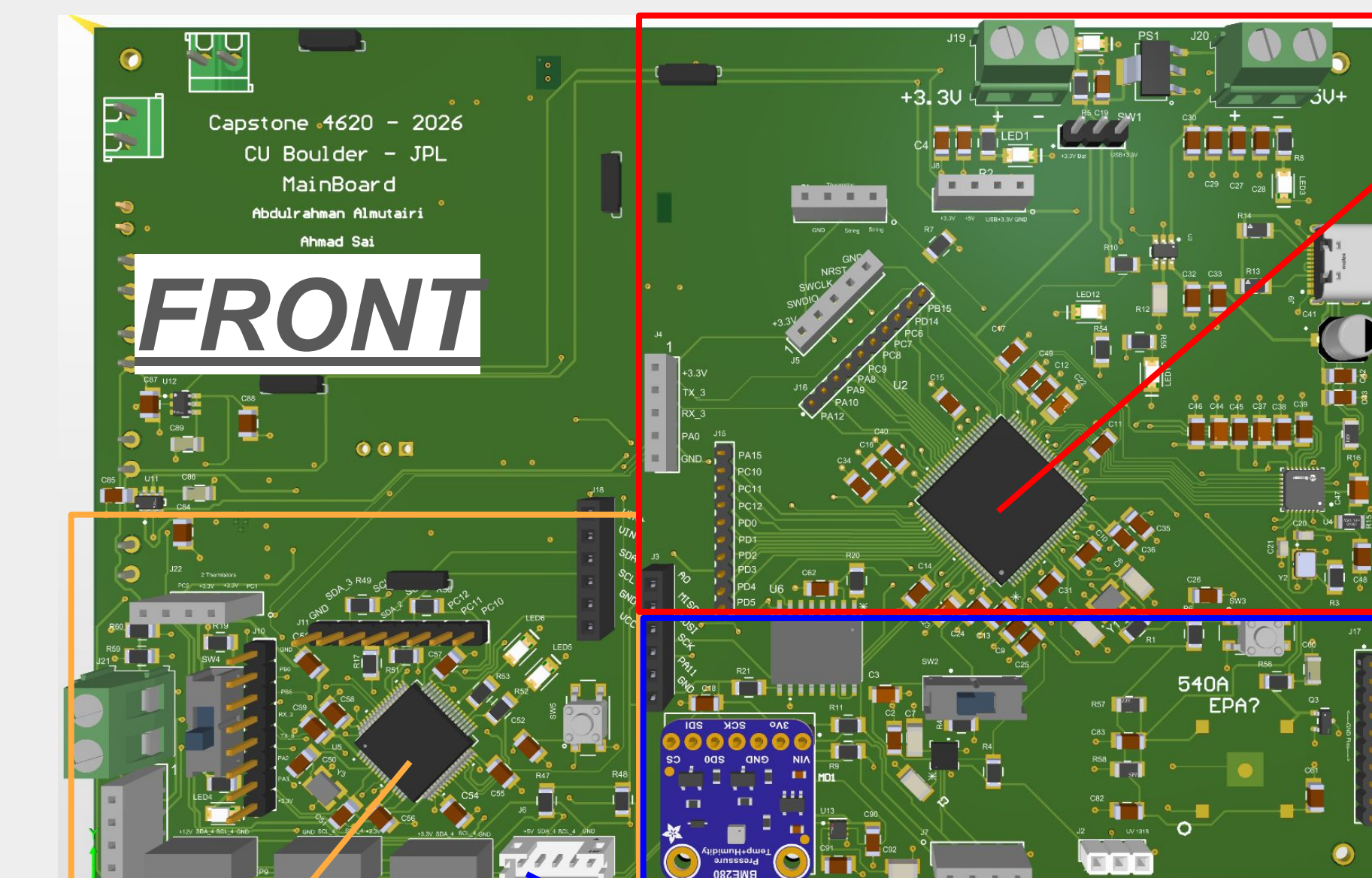


Radio Communication



- 720Kbps Transmission at 430Mhz
- BPSK modulation with RRC Filtering to minimize signal interference
- Transmit diagnostics and measurements on monthly schedule

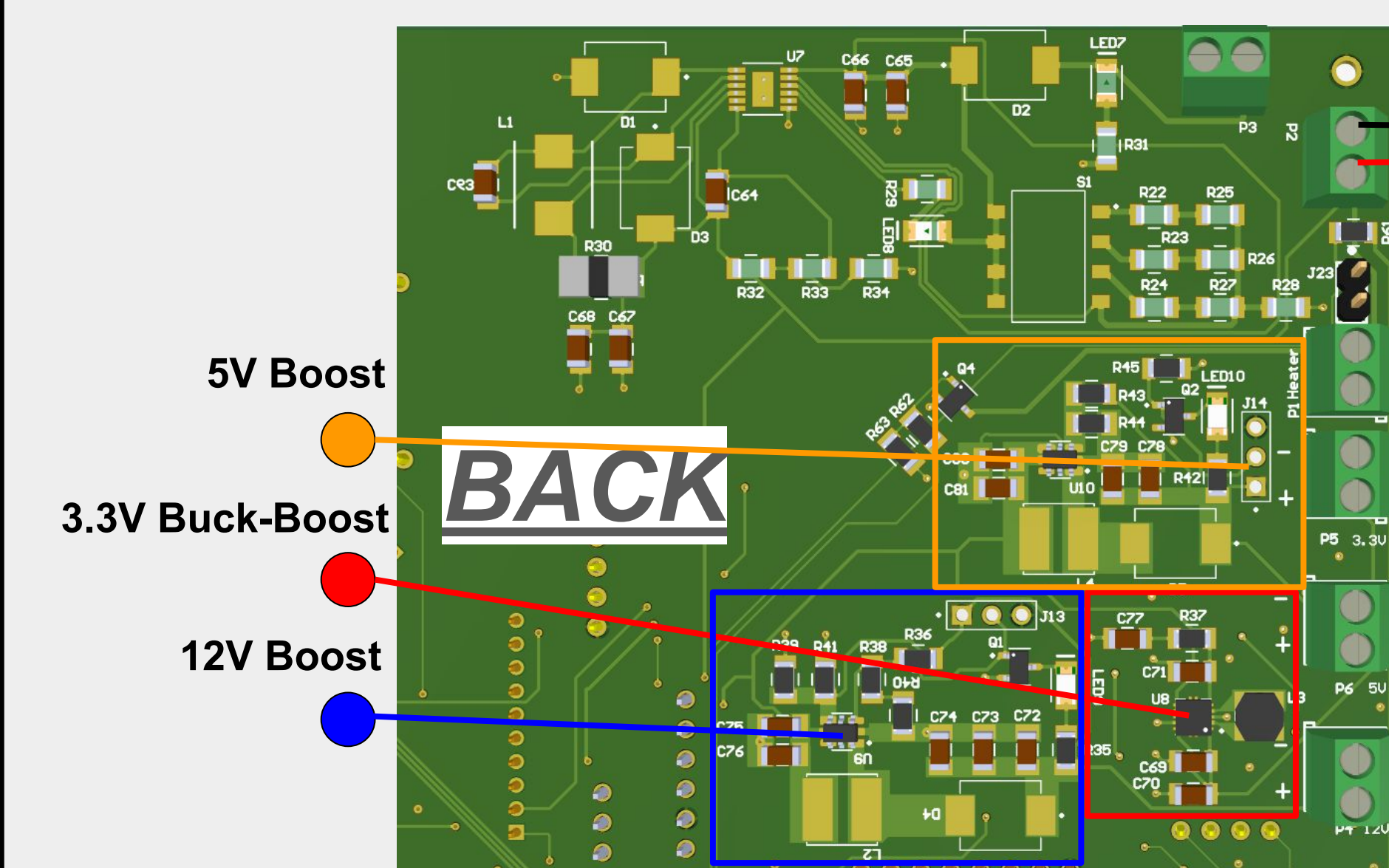
Mainboard and Batteries



- High Power STM32 H7
- Processing Commands
- Controlling Sensors
- High Speed USB
- Managing Payloads and Sensors

- Low Power - STM32L4 MCU
- Scheduling Tasks
- Power State manager
- Heater Control

- On Board Sensors and Payload Ports
- Payloads Support - I2C:
 - Type I: 3.3v
 - Type II: 5v
 - Type III: 12v



- Battery
 - 18650 platform
 - 1S9P 3.7V battery pack
 - 116 Whr capacity

Software

- L4 (Low-Power Controller)
 - Schedules system tasks
 - Turns the system on, off, or into low-power mode
 - Sends commands to H7 to execute
- H7 (High Performance Processor)
 - Executes sensing, storage, and transmission tasks
 - Activates only when needed to save power
 - Interfaces with storage and radio devices for high speed data transfer